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Enhancing Well Integrity with Managed Pressure Drilling and Cementing in Challenging Zones with Significant Fluid Losses

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Abstract

The project, conducted in challenging drilling environment, faced significant operational challenges, including high risks of fluid losses and difficulties in running and cementing liners. This paper examines the application of Managed Pressure Drilling (MPD) technology to address these challenges, emphasizing its role in maintaining and enhancing cementing operations. The implementation of MPD allowed for precise control of annular pressure, significantly mitigating the risks associated with fluid losses and ensuring a smoother delivery of the section.

MPD technology was instrumental in optimizing liner running and cementing operations by stabilizing wellbore conditions and reducing the likelihood of formation fractures during cementing. The ability of MPD to maintain consistent pressure profiles improved the integrity of cement bonds and achieved effective zonal isolation. The data collected from the project demonstrated a noticeable improvement in operational efficiency, with reduced non-productive time and enhanced safety measures, confirming MPD's effectiveness in such demanding scenarios.

The paper underscores the substantial benefits of deploying MPD technology in drilling operations within complex geological formations. By achieving a significant reduction in drilling fluid losses and enhancing the integrity of liner cementing processes, MPD proves its efficacy in converting challenging drilling scenarios into successful operations. This paper provides critical insights and advanced methodologies for cementing with MPD, advocating for its broader implementation in similar drilling contexts to optimize project outcomes and bolster operational reliability.

Introduction

Drilling operations frequently encounter significant challenges, especially when navigating narrow drilling windows and managing fluid losses. This project utilized MPD to address these challenges and ensure efficient and safe operations. Initially, the well Experienced substantial fluid losses in formation A, characterized by a lower pressure compared to the higher pressure in formation B. This situation posed risks of fluid losses and potential cross-flow, complicating the drilling process. MPD was implemented to stabilize the wellbore, mitigate losses, and manage the equivalent mud weight (EMW) effectively. Key

objectives of MPD included rapid adjustments to bottom hole pressure (BHP) to maintain control and minimize risks of kicks and losses. The ability to determine true formation pressure through static pore pressure tests (PPT) was crucial for refining drilling strategies and maintaining safe operations. MPD operations began with an equivalent circulating density (ECD) of 94 pcf, using a mud weight (MW) of 88 pcf. This setup allowed for necessary adjustments to prevent further gain or loss scenarios. Adjustments in BHECD countered flow tendencies, ensuring wellbore stability. Despite these measures, complete fluid losses occurred, necessitating immediate remedial actions such as loss circulation material (LCM) pills and poly plugs to reinforce formation A. These interventions restored full surface returns and enabled an increase in MW, although formation B continued to pose challenges to operations due to higher pressures. Strategic adjustments in ECD maintained stability without kicks, allowing smooth progression to the target depth. The differential pressures and instability of the formations required precise pressure management to maintain the well integrity. MPD technology enabled effective control over wellbore conditions, mitigating associated risks. Given these challenges, achieving successful zonal isolation was difficult. This paper focuses on the cementing techniques used, highlighting strategies and innovations to overcome these challenges and ensure well integrity.

Adopting a proactive approach to loss management

Losses were initially observed in Formation A and subsequently propagated into Formation B, bringing the operation close to the limits of the available drilling window. MPD was employed to dynamically manage the ECD profile and identify an optimal SBP that would minimize losses, prevent influx, and maintain wellbore stability. However, the pressure required to stabilize Formation B exceeded the fracture gradient of Formation A, necessitating reinforcement of the upper formation. To address this, LCM was pumped, and a loss control pill was displaced to the bit and squeezed into the loss zone under controlled SBP. Through rapid SBP adjustments and precise ECD control, a stable drilling margin was successfully reestablished, enabling safe and continuous drilling operations.

During the ongoing efforts to mitigate fluid losses, gas was detected at the surface, prompting the need for an immediate well shut-in to ensure safety and maintain control. The shut-in procedure was executed against the RCD, allowing the drilling team to maintain rotation of the drill string while safely circulating the gas out of the wellbore. This method ensured that the well remained under control without interrupting the circulation process, which was critical for maintaining pressure balance and avoiding further complications.

Throughout the gas handling operation, careful attention was given to maintaining a consistent and accurate pressure profile within the wellbore. This was essential to prevent additional influxes or losses while the gas was being circulated out. At the same time, LCM was continuously pumped into the well to reinforce the integrity of the upper formation, known as Formation A. Strengthening this formation was necessary to enable it to withstand the higher pressures required to maintain control over the deeper and more pressurized Formation B. The coordinated execution of gas circulation and loss treatment allowed the operation to proceed safely and efficiently, preserving well integrity and enabling continued progress toward the target depth.

Figure 1 illustrates how rapid adjustments to SBP were made in response to dynamic well conditions. By closely monitoring pumped volumes and managing the crossflow between Formation A and Formation B, the team was able to maintain the well balance and identify the optimal ECD required for stable operations.

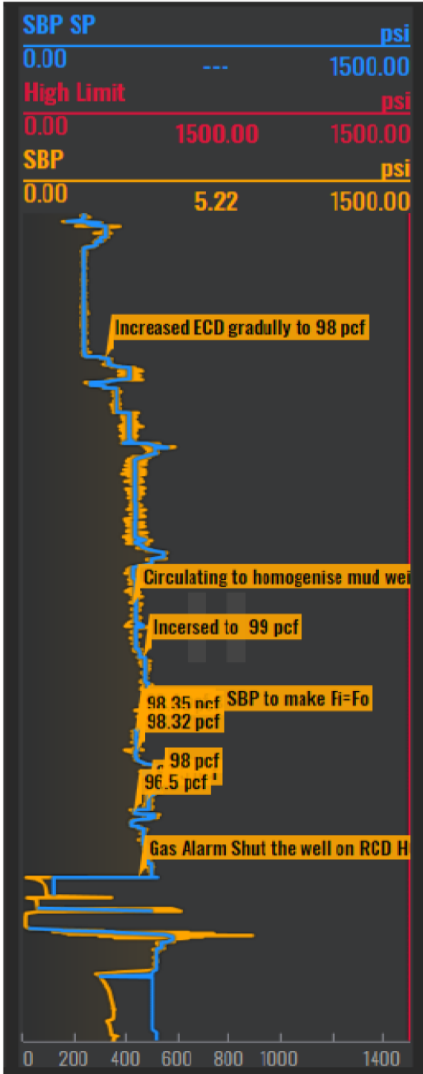


Figure 1—Complications at the beginning of operations

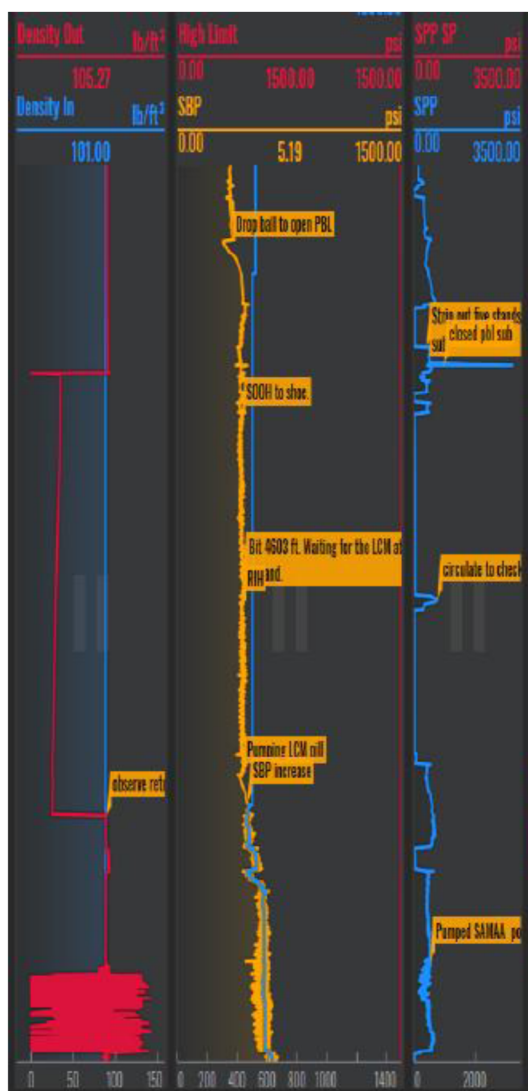


Figure 2—Well Shut in against RCD

Following the successful treatment of fluid losses and reinforcement of the upper formation, drilling operations resumed with significantly improved stability. The application of MPD continued to play a critical role in maintaining well control and operational efficiency. With the loss zones effectively sealed and the pressure profile stabilized, drilling progressed without further losses. Throughout this phase, a consistent and controlled pressure environment was maintained by sustaining a compulsory ECD of 100.5 pcf. This precise pressure management ensured that the well remained balanced, preventing both influx and additional losses, and allowed for uninterrupted advancement toward the planned total depth.

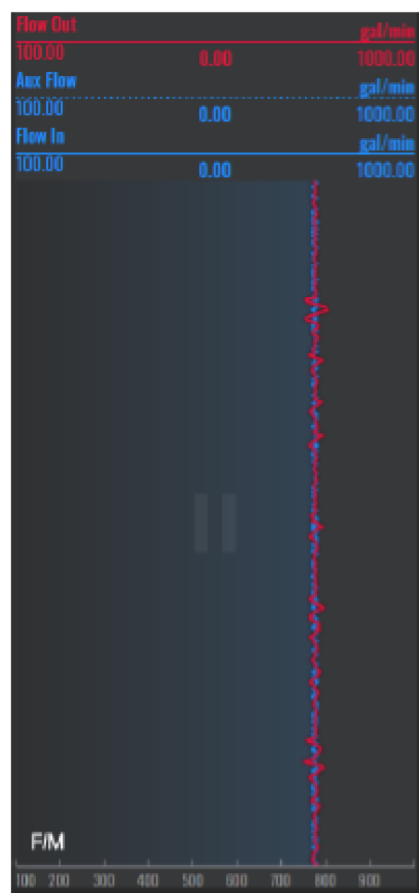


Figure 3—Drilling with MPD with full returns

Although MPD enabled precise control over wellbore pressure and allowed for stable drilling at an ECD of 100.5 pcf, transitioning to the next operational phase, pulling out of the hole in preparation for liner running, required a deliberate shift in pressure management strategy. While MPD maintained the well in a balanced state during drilling, tripping operations necessitated the establishment of an overbalanced condition to ensure well control in a static environment. This was achieved by spotting a kill mud weight heavier than the circulating density used during drilling, thereby exceeding the 100.5 pcf threshold.

However, this increase in MW reintroduced the risk of fluid losses, particularly in the previously treated weak zones of Formation A. The well, which had been stabilized under MPD conditions, now had to contend once again with the potential for induced losses due to the higher hydrostatic pressure. Despite this, the operation could not proceed without creating a safe overbalance, especially given the need to maintain well integrity during the pulling out to pick up the liner.

To mitigate these risks, the rig team proactively prepared for the anticipated losses by ensuring sufficient volumes of kill mud and loss circulation materials were available on location. This readiness allowed for a controlled response to any losses that occurred during the transition. The MPD system remained on standby to assist in pressure management if needed, and operational parameters were closely monitored throughout the process.

Once the well was conditioned with the heavier mud and stabilized under the new pressure regime, the operation advanced to running the liner. This phase required careful coordination to maintain wellbore stability, manage surge and swab effects, and ensure that the liner could be deployed to the bottom without compromising the integrity of the well.

Running Liner

As the operation progressed to the liner deployment phase, the well continued to exhibit challenges due to persistent fluid losses. Upon initiating the run-in-hole with the liner, losses were immediately observed as the assembly entered the open hole section. These losses intensified with each additional stand, ultimately resulting in a complete loss of returns. This indicated that the formation could no longer support the hydrostatic pressure of the drilling fluid, despite previous mitigation efforts.

During the liner operation, significant fluid losses were encountered shortly after setting the bearing assembly. Notable discrepancies in surface volumes were observed during the running of two stands, indicating considerable losses. As an initial countermeasure, the tripping speed was reduced to 7 minutes per stand to mitigate the surge effect on the formation. Despite this adjustment, losses persisted, prompting a dynamic flow check at a controlled rate of 3.5 bpm, which revealed that only 70% of the circulation was returning to the surface, emphasizing the severity of the losses.

The original displacement plan, which involved transitioning from 101 pcf kill mud to 94 pcf mud at the bottom hole, was rendered unfeasible due to ongoing losses. Displacement from heavy mud to lighter mud was initiated at the current depth, unexpectedly resulting in a total loss scenario. The weak part of the formation was unable to withstand the pressure exerted by the 101 pcf mud, which compounded the issue. Additionally, as the liner was being run into the hole, a surge effect occurred, exacerbating fluid losses. These factors contributed to the total loss scenario, creating significant challenges. Despite these difficulties, liner installation continued, with the well being filled through an auxiliary pump using 101 pcf kill mud to maintain well integrity. The SBP remained at 0 psi, indicating that total loss conditions persisted and highlighting the need for careful pressure management to mitigate further risks. By continuously filling the well while it remained shut in on RCD, the operation maintained a high level of safety. This approach ensured that any unexpected influx reaching the surface would be contained within the closed system, eliminating the risk of uncontrolled flow. Importantly, this safety measure was implemented without interrupting the ongoing liner deployment, allowing operations to proceed without delay.

The liner used in this section was specifically selected for its compatibility with MPD operations, which enabled precise pressure control throughout the deployment process. This compatibility proved essential in maintaining well control and ensuring the liner could be safely landed at the intended depth.

The ability to run the liner under total loss conditions, while maintaining a full annulus and stable pressure profile, was a critical achievement. It not only preserved the integrity of the wellbore but also set the stage for the next major challenge—executing the cementing operation under similarly demanding conditions. With the liner successfully in place and the MPD system continuing to provide real-time pressure management, the operation transitioned into the cementing phase, where maintaining wellbore stability would remain paramount.

Cementing Operations

Upon reaching the intended liner depth, the operation transitioned to using a 94 pcf MW to displace the annulus up to the presumed loss zone. Simultaneously, 94 pcf mud was circulated through the drill string to displace both the internal drill string volume and the annular volume from the bottom of the well up to the loss zone. This dual-displacement strategy was carefully designed to maintain well control, minimize pressure fluctuations, and reduce the risk of further destabilizing the formation, and at the same time the well is being shut on RCD. At the conclusion of this displacement, surface back pressure SBP stabilized at 180 psi, indicating that total losses would occur at an equivalent static density (ESD) of approximately 97.7 pcf at the loss zone.

Recognizing the limitations imposed by the loss-prone formation, the team developed a revised cementing strategy explicitly tailored for total loss conditions. A key component of this strategy was to maintain a closed choke configuration until the SBP naturally built up to, but did not exceed, 200 psi.

This pressure level was interpreted as the point at which the well was balanced with an EMW of 97 pcf. Maintaining this balance was critical to prevent the ECD from falling below the threshold required to control the formation at the anchor point. The strategy emphasized maintaining equilibrium between flow-in and flow-out rates to avoid inducing further losses or risking influx.

As the cementing process progressed and the top of the lead slurry approached the critical interval, a gradual increase in SBP was observed. In response, the MPD choke was actively adjusted to maintain the SBP at 200 psi. This adjustment allowed for full returns at the surface, confirming that the well remained in a balanced state. Once full returns were established, the SBP was carefully reduced in a controlled manner, while continuing to balance the inflow and outflow, to ensure that the pressure profile remained within the safe operating window.

Toward the end of the displacement, a reduction in returns was observed, with surface returns decreasing to approximately 73%. This indicated that some fluid was still being lost to the formation, although the well remained under control. Prior to isolating the annulus, a brief Coriolis flow check was conducted using the MPD system to verify that the well was statically balanced. This check confirmed that the well remained in a safe condition, allowing the operation to proceed to the next step.

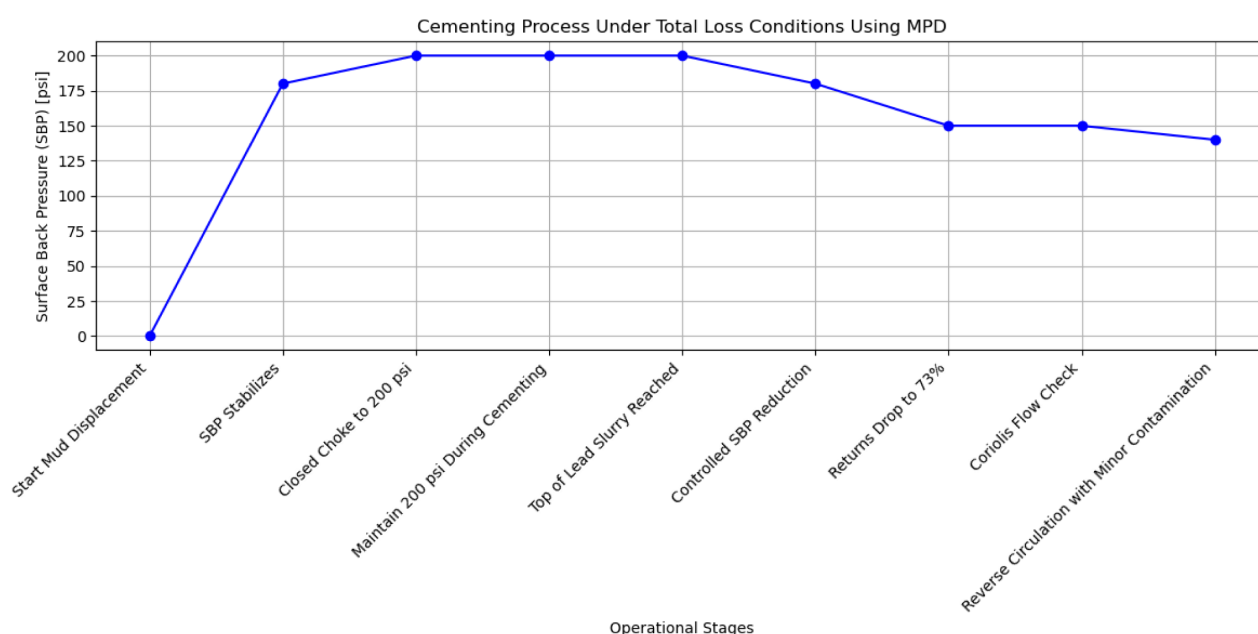


Figure 4—SBP profile during cementing the liner

During reverse circulation, cement contamination was observed at the surface, indicating that cement was placed. While this presented a minor operational challenge, it did not compromise the overall integrity of the cementing job. The ability to dynamically adjust operational parameters in real time, maintain pressure control, and respond to evolving downhole conditions was pivotal to the success of this operation.

This scenario underscores the exceptional complexity of conducting cementing operations under total loss conditions and highlights the indispensable role of MPD in maintaining wellbore stability and precise pressure control. Unlike conventional cementing operations, which typically rely on predictable formation behavior and static pressure environments, this case demanded a highly adaptive and responsive approach. The integration of MPD into the cementing workflow transformed what would have been a high-risk, potentially compromised operation into a controlled and successful execution.

The cementing process in this case was far from routine. It required continuous real-time monitoring and adjustment of SBP to respond to evolving downhole conditions. The team had to dynamically manage ECD to avoid further losses while ensuring that the cement slurry was placed effectively to achieve zonal

isolation. This level of operational agility is not characteristic of traditional cementing practices and reflects a paradigm shift in how high-risk wells are managed.

Moreover, the success of this operation was not solely the result of advanced pressure control technology. It was equally a testament to meticulous pre-job planning, cross-disciplinary collaboration, and operational discipline. The team's ability to anticipate potential complications, prepare contingency volumes of mud and cement, and coordinate seamlessly between drilling, MPD, and cementing crews was instrumental in achieving the desired outcome. Every stage of the operation—from liner running under total loss conditions to the final cement displacement—required a tailored response based on real-time data and formation feedback.

This case reinforces the value of MPD not only as a drilling enhancement tool but as a critical enabler of complex well construction operations in high-risk environments. It demonstrates that with the right integration of technology, planning, and execution, even the most unconventional and challenging cementing scenarios can be navigated successfully.

As shown in Figure 5, the cementing operation began while the well was under total loss conditions. The well was shut in on the MPD chokes to monitor pressure buildup, which indicated that returns were reaching the surface. This is illustrated on the right side of the figure by the green line representing Choke A. SBP, shown by the yellow line, initially built up to 180 psi, and then further increased to 200 psi.

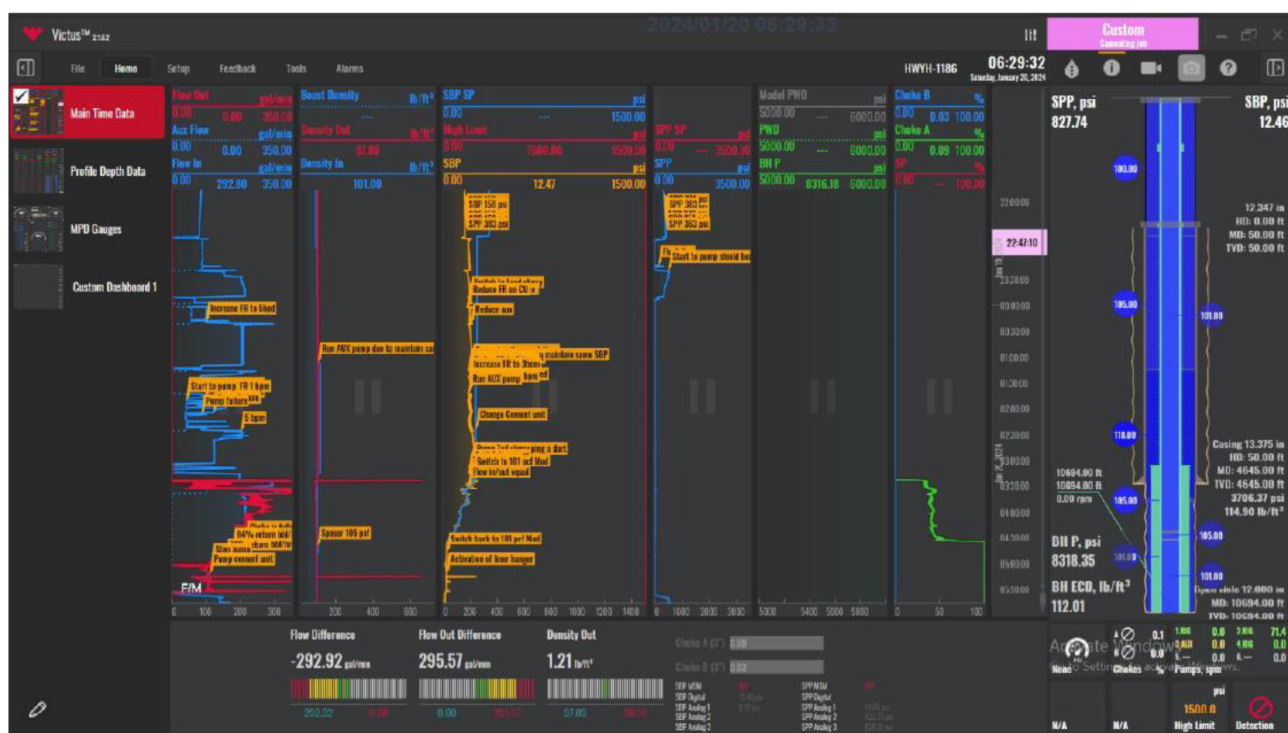


Figure 5—Cementing with total losses

At this point, the choke was gradually opened to maintain a controlled balance between flow-in and flow-out rates, allowing the cementing operation to continue under stable conditions. This pressure behavior confirmed that the cement had reached the loss zone and zonal isolation had commenced. The subsequent pressure increase on the choke indicated continued cement placement. To maintain a constant ECD at this critical stage, the choke was adjusted accordingly, and the SBP was sustained throughout the remainder of the operation.

Conclusion

This case study highlights the critical role of MPD in enabling successful well construction in one of the most technically demanding environments, characterized by severe fluid losses and the absence of a viable drilling window. MPD allowed for precise control of the wellbore pressure profile, enabling the drilling team to dynamically manage the ECD and maintain well integrity throughout the drilling phase. By maintaining a narrow pressure margin at 100.5 pcf, the operations progressed without further losses after initial mitigation efforts, demonstrating the effectiveness of MPD in stabilizing complex formations.

However, transitioning from drilling to tripping operations introduced new challenges. To safely pull out of the hole in preparation for liner running, it was necessary to establish an overbalanced condition by spotting a kill mud weight that exceeded the previously maintained circulating density. This deliberate increase in hydrostatic pressure reintroduced the risk of losses, particularly in the previously treated weak zones of Formation A. Despite this, the operation could not proceed without ensuring well control during the static phase. The rig was fully prepared for this scenario, with sufficient volumes of kill mud and loss circulation materials staged to manage the anticipated losses and maintain operational safety.

Once the well was conditioned with the heavier mud and stabilized, the operation advanced to running the liner. This phase required careful coordination to manage surge and swab effects, maintain wellbore stability, and ensure the liner could be deployed to the bottom without compromising the integrity of the well. MPD remained on standby to assist with pressure control if needed, and operational parameters were closely monitored throughout the process.

The cementing phase presented its own set of challenges, particularly under total loss conditions. A key innovation was the real-time adjustment of SBP using MPD. By shutting in the well on MPD chokes and gradually filling the annulus with MPD mud, pressure was built up to a controlled level just below the fracture gradient. This pressure was then maintained consistently during cement displacement by carefully managing flow-in and flow-out rates, ensuring full returns and preventing further losses. The ability to monitor and respond to downhole conditions in real time was instrumental in overcoming total losses encountered during liner running and in achieving a successful cement job.

Additionally, during the loss mitigation phase, gas was observed at the surface, requiring a controlled shut-in against the Rotating Control Device (RCD). This allowed the team to maintain rotation while safely circulating the gas out of the wellbore. Simultaneously, lost circulation material was pumped to reinforce the upper formation, enabling it to withstand the higher pressures needed to control the deeper formation. This coordinated response ensured both well control and continued progress in loss treatment operations.

Ultimately, this operation underscores MPD's value not only as a drilling enhancement tool but as a comprehensive pressure management system. Its ability to adapt to dynamic downhole conditions, manage complex pressure regimes, and support critical transitions such as tripping and cementing makes it an indispensable technology in high-risk, loss-prone environments. The methodologies and lessons presented in this paper offer a valuable reference for future operations in similarly complex geological settings.

Nomenclature

ECD	Equivalent Circulating Density, pcf
EMW	Equivalent Mud Weight, pcf
ESD	Equivalent Static Density, pcf
LCM	Lost Circulation Material
MPD	Managed Pressure Drilling
MW	Mud Weight, pcf
PPT	Pore Pressure Test
RCD	Rotating Control Device
SBP	Surface Back Pressure, psi